MID-TERM EXAMINATION
(Course Name : B. Tech(AI&ML)) (Semester : IV)
(March, 2024) OFF LINE mode

Subject Code BAM-202	**Subject:** Machine Learning
Time : 1 ½Hours	**Maximum Marks :** 30
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) What are the primary goals of machine learning, and how do they differ from traditional programming approaches?	UNIT-1	
	(b) Use min-max and Z -transform normalization methods to *normalize* the following group of data: 200, 300, 400, 600,1000	UNIT-1	
	(c)What is the predicted probability that a single 20-year-old female with an annual household income of Rs 10,000 responded to email campaign. (Gender=1 coded for male and 0 for female , Married=1 for married and 0 for unmarried) (Use the following coefficients)	UNIT-2	

Terms	Estimate
constant	-0.682
Gender 0	1 -0.7779
Married 0	0.17
Income 10,000	0.0003
Age 20	-0.228

	(d) In a LASSO regression. What happens to the estimated regression coefficients if the regularization parameter λ is very high?	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1**		**(5,5)**
	(a) Explain the importance of data visualization in data exploration and preprocessing. What are some commonly used visualization techniques, and how do they aid in identifying patterns and relationships in data?		
	(b) Explain the importance of data preprocessing in machine learning? How does data preprocessing contribute to improving the performance and reliability of machine learning models?		
	(C)Explain the concept of normalization and its significance in the context of feature scaling.		
Q3	**(Attempt any Two Parts) UNIT-2**		**(5,5)**
	(a) You are provided with a dataset containing information about the number of hours students spend studying and their corresponding scores on a test. You plan to use linear regression to predict the test score based on the number of study hours.		

Consider the following subset of the dataset:

Study Hours (x)	Test Score (y)
2	65
3	70
5	75
7	82
8	88

	Using linear regression, determine the equation of the regression line that best fits the data. Then, predict the test score for a student who studies for 6 hours. Compute R square?
	(b)Consider a dataset with the following two-dimensional points for a binary classification problem: Class 1 (Positive class): Point A: (2, 3) Point B: (4, 5) Point C: (3, 1) Class 2 (Negative class): Point D: (6, 5) Point E: (7, 7) Point F: (8, 3) You are tasked with implementing a k-NN classifier with k=3 and classifying a new data point G at coordinates (5, 4). Use the Euclidean distance metric for the calculations. Explain why choosing a different value for k (e.g., k = 5) might affect the prediction and how it could potentially impact the overall performance of the KNN model.
	(c) Compare and contrast Support Vector Machines with Logistic Regression for binary classification tasks. What are the strengths and weaknesses of each algorithm, and under what circumstances might one be preferred over the other?